

Exercise 10

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

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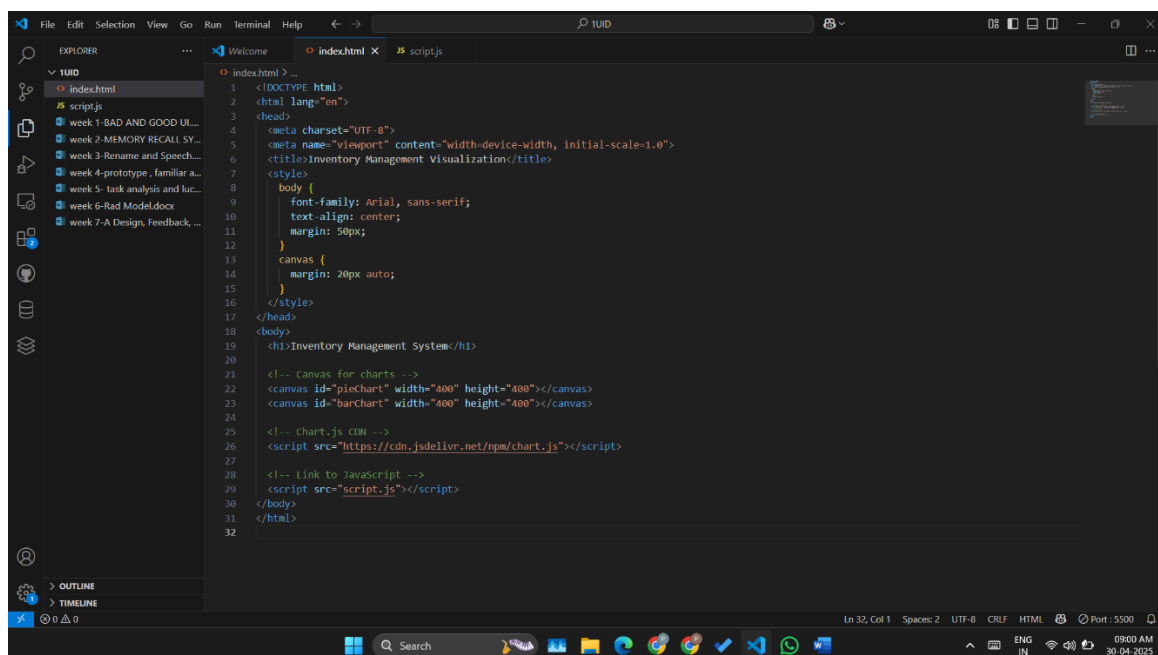
Aim:

To create simple charts (pie and bar) using JavaScript and Chart.js to show inventory data clearly.

Step-by-Step Process

Step 1: Set Up HTML File

- Created an HTML file named index.html.
- Added a heading to show the title "Inventory Management System".
- Used two <canvas> elements to display the pie chart and bar chart.
- Added basic CSS to make it look neat and centered.
- Included the Chart.js library using a CDN link.
- Linked a separate JavaScript file called script.js.



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Inventory Management Visualization</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      text-align: center;
11      margin: 50px;
12    }
13    canvas {
14      margin: 20px auto;
15    }
16  </style>
17 </head>
18 <body>
19   <h1>Inventory Management System</h1>
20
21   <!-- Canvas for charts -->
22   <canvas id="pieChart" width="400" height="400"></canvas>
23   <canvas id="barChart" width="400" height="400"></canvas>
24
25   <!-- Chart.js CDN -->
26   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
27
28   <!-- Link to JavaScript -->
29   <script src="script.js"></script>
30 </body>
31 </html>
32
```

Step 2: Write JavaScript Code for Charts

- Created a JavaScript file named script.js.
- Defined the inventory data with item categories like Electronics, Clothing, etc.
- Added values for the number of items in each category.
- Assigned different colors to each category for better understanding.
- Used Chart.js to draw:

- o A pie chart to show the percentage of each category.
- o A bar chart to show the count of items in each category.

```

1 // Inventory data
2 const inventoryData = {
3   labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
4   datasets: [{
5     label: 'Items in stock',
6     data: [200, 150, 100, 80, 50],
7     backgroundColor: [
8       '#FF6384', // Electronics
9       '#36A2EB', // Clothing
10      '#FFCE56', // Home Appliances
11      '#4BC0C0', // Books
12      '#9966FF' // Toys
13    ]
14  }]
15 };
16
17 // Pie Chart
18 const ctxPie = document.getElementById('pieChart').getContext('2d');
19 new Chart(ctxPie, {
20   type: 'pie',
21   data: inventoryData,
22   options: {
23     responsive: true,
24     plugins: {
25       title: {
26         display: true,
27         text: 'Inventory Distribution'
28       }
29     }
30   }
31 });
32
33 // Bar Chart
34 const ctxBar = document.getElementById('barChart').getContext('2d');
35 new Chart(ctxBar, {
36   type: 'bar',
37   data: inventoryData,

```

Output

- The browser displayed:
 - o A pie chart that shows how inventory is divided.
 - o A bar chart that shows item quantities in each category.
- Charts are clear and easy to understand.

